J.A.R.V.I.S BLUEPRINT

SYSTEM ARCHITECTURE & TECHNOLOGY STACK SPECIFICATIONS

DOCUMENT ID: AP-J2-9842
VERSION: 2.1.0
CLASSIFICATION: PUBLIC

Executive Overview: J.A.R.V.I.S is an advanced local-first agentic AI operating system combining high-speed wake-word speech processing, dual-brain LLM reasoning, automation engines (browser, email, RAG, and email agents), and real-time state sync with a customized HUD console interface.

CORE SYSTEMS & PROGRAMMING STACK

Programming Languages

- **Python 3.10+**: Core logic runtime, agent control loop, background threads.
- **HTML5 / CSS3 / JavaScript (ES6)**: HUD web UI, real-time telemetry gauges, Arc Reactor SVG canvas animations.

Data Persistence & Memory

- **SQLite**: Default local relational memory structure.
- **Pyperclip Clipboard Buffer**: System-wide clipboard sharing and RAG response backup memory.

AUTONOMOUS AGENTS & CONTROLLERS

Voice Engine (STT & TTS)

- **Microphone Listener**: Speed-calibrated speech-recognition buffer.
- **Wake Word Detector**: Fast multi-phrase intent detection matching "Jarvis" variants.
- **Audio Queue (TTS)**: Asynchronous Pygame mixer queue playing voice notifications.

Automations & Integrations

- **Browser Agent**: Headless Playwright script handler.
- **Email Agent**: Pre-filled direct compose Gmail dispatcher.
- **Google Workspace Hook**: Apps Script bridge to Gmail, Drive, Calendar, Docs.

J.A.R.V.I.S BLUEPRINT

LIBRARIES, DEPENDENCIES & INTEGRATION MAP

EXTERNAL TOOLING & MODELS LAYER

Local Reasoning Models

- **Ollama**: Orchestrates execution of offline LLMs.
- **Qwen 2.5 (1.5B/4B)**: Primary/secondary task classification.
- **Mistral 7B**: Large fallback reasoning tasks.

Service Hooks

- **n8n Webhooks**: Web API workflow trigger.
- **Telegram Bot API**: Remote telemetry bot handler.
- **Flask / Waitress**: Local telemetry stats REST API.

KEY PYTHON LIBRARIES & PACKAGES

PACKAGE NAME	CATEGORY	ROLE IN J.A.R.V.I.S
`pyautogui`	OS Automation	Automates clicks, keystrokes, and UI interactions across desktop apps.
`pyperclip`	System Clipboard	Stores latest generated LLM responses to clipboard for quick paste/emailing.
`pygame`	Audio Mixer	Handles sound effects and provides non-blocking playback queue for TTS.
`win32gui / win32con`	Windows OS API	Monitors active foreground window titles for context-aware RAG operations.
`speech_recognition`	Speech Logic	Performs microphone speech-to-text processing via local/remote models.
`flask / waitress`	REST API	Serves status JSON queries to the HUD web client console loop.
`beautifulsoup4`	Data Parsing	Extracts key clean text structures from web pages and documents.

JARVIS-SYSTEM-SPECIFICATIONS • GENERATED VIA HEADLESS CHROME VECTOR PRINT ENGINE • CONFIDENTIALITY: NONE